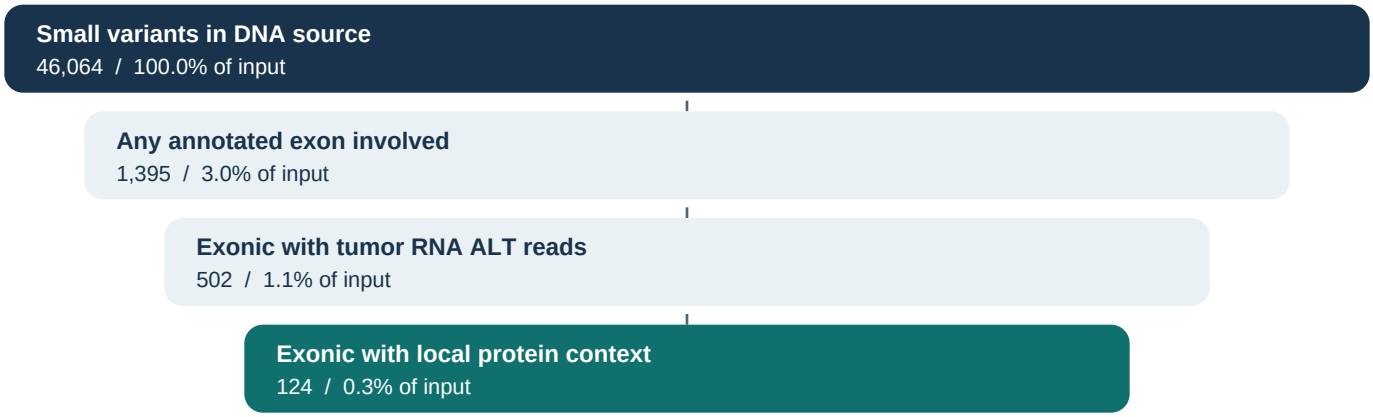
COLO829: somatic mutation and RNA evidence

Whole source-call audit | matched-normal DNA baseline | original tumor RNA | pinned reference ORFs. Counts below describe the specified public call sets and RNA run, not every biological mutation or a validated full-length isoform.

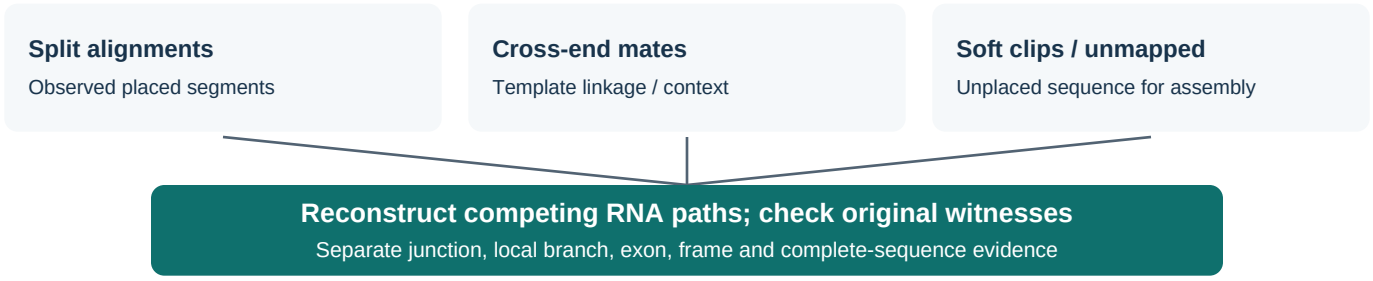
Small variants: DNA to RNA and local protein



Boxes show subset order; widths are schematic, not scaled to count.

Structural variants: complementary routes to RNA reconstruction

188 nominated SV records; 106 have RNA near an end



The legacy split/mate screen is one route, not an eligibility gate.

Soft-clip counts were not enumerated in the old atlas; this channel is not zero.

Breakend-window coverage is locus coverage. Split reads, cross-end mates and soft clips are complementary discovery evidence; none alone resolves a coding frame. Exon overlap is an annotation, not an eligibility filter. The older exon-restricted split/mate screen and selected junction replay are documented later without treating their negatives as absence. The newer COLO829 workflow uses all retained paths and exon-anchored clips.

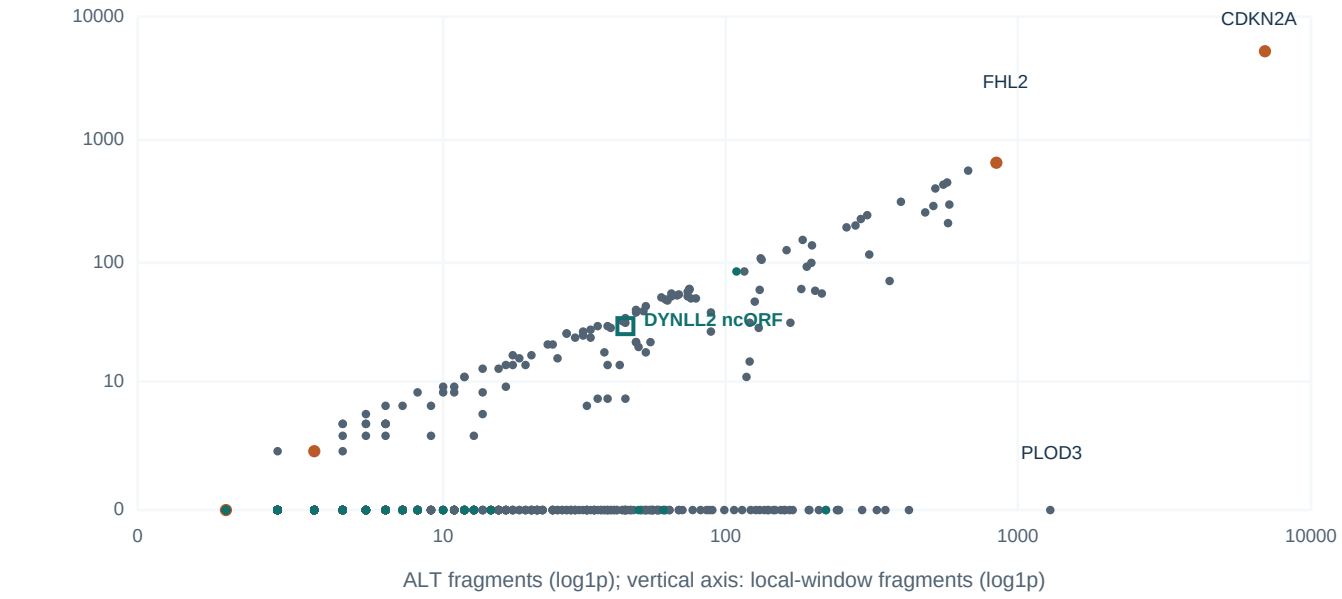
Variant landscape: depth, coding recovery and unresolved categories

Every nominated small variant is counted above. Dots show original-atlas outcomes: baseline dots had no local protein context under those settings. The labelled open square adds the new default-coverage ncORF result where present; original counts stay unchanged. A context may be reference-identical. Cultures and assays remain separate.

DNA inventory composition and original-RNA outcomes

	DNA rows	ALT calls	Atlas context
SNV	44,005	1,865	120
insertion	961	39	2
deletion	1,098	29	2

RNA allele support versus support for the top local protein window



Category	What this report can establish
Small variants	Original-RNA allele observations and bounded local coding windows
SVs / fusions	Original joins and competing RNA paths; annotated coding frame only where resolved
ncORFs	Validated reference ORF frames and mutation-supporting RNA; candidate translation remains conditional
Viral / other antigen sources	Not surveyed in these host-focused inputs; no absence claim

At a glance

Evidence layer	Count	Interpretation
Ingested small variants	46,064	44,005 SNVs; 2,059 indels
Ingested SV source records	188	DNA breakpoint records/events, counted in their source sets
SVs involving an annotated exon	97	Span for DEL/DUP/INV; nominated breakpoint or insertion flanks otherwise
Any annotated exon involved	1,395	At least one changed base or insertion flank overlaps an Ensembl 87 exon
Ensembl 87 coding effect	373	Any annotated protein-coding transcript, reference model
orfdata ncORF coding-block overlap	5	Candidate reference ORF overlap; distinct from mutant translation
Original RNA ALT-observed calls	1,933	35,949 ALT reads; 35,617 ALT fragments
Atlas local protein contexts	124	12,218 ranked contexts; new ncORF pass reported separately

Coding annotation and its denominator

Of 46,064 DNA small-variant records, 373 change the coding sequence of at least one Ensembl 87 transcript. The pinned orfdata 0.34.1 GRCh38 catalog (7,264 reference ncORF models; release 2026-10-11.1) has exact coding-block overlaps at 5 calls. The groups overlap at 0 calls, so the union is 378. These are annotation opportunities. The catalog does not assess a cell-line-specific mutant ORF, RNA expression, or peptide evidence.

Reference annotation	All DNA calls	With original RNA ALT
Ensembl coding effect	373	180
orfdata coding-block overlap	5	2
Union of either category	378	see per-call table

SNVs query their changed reference base. Deletions query removed reference bases; insertions use the VCF anchor as a candidate overlap. The per-call table preserves that rule. A coding-block hit does not prove the inserted sequence is in frame.

Any annotated exon is involved in 1,395 calls (3.0% of the small-variant source set). This includes UTR and noncoding-transcript exons. Deletions query every removed reference base, including exon crossings with intronic endpoints; insertions query both neighboring reference bases as candidate exon involvement.

Original RNA and local protein evidence

RNA assay: SRR8615617, Illumina paired RNA. Each nominated small variant has a status from this run. An ALT read is a sequence observation; it is not a full-length mutant isoform.

RNA allele status	Small-variant records
alternate observed	1,933
reference only	1,637
reference and other only	36
other allele only	27
alignment without allele call	166
no alignment overlap	42,265

The 1,933 ALT-observed calls contribute 35,949 allele-read and 35,617 fragment observations. Isovar resolves at least one read-compatible **local** protein context for 124 calls (12,218 ranked rows). The remaining 1,809 ALT-observed calls have no local translation under the stated method. No-overlap and uncallable statuses cannot establish absence of RNA expression.

At CDKN2A chr9:21971154:CCG>C, all 6,984 ALT allele reads were evaluated. The top local frameshift context is **MMMGSARVAELLLLHGGAQLRRPR** with 5,338 supporting reads. Its 1,416 ranked local contexts reflect read-spanning alternatives, not full-length isoforms. TAGET polished representatives are an independent culture and assay; their cluster counts do not measure mutant allele support.

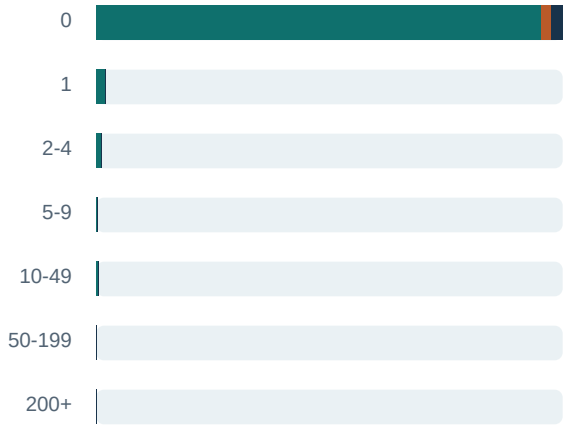
Structural-variant RNA screen

Breakend screen outcome	Source records
No breakend-window alignment	78
One end covered; no split candidate	36
Both ends covered; no split candidate	57
Cross-end mate context; path unresolved	11
Partner coordinate unresolved	6

The two SV benchmarks remain separate; no junction or SV-derived protein is called.

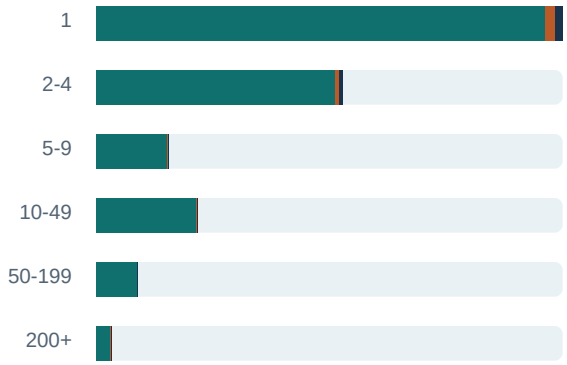
RNA alternate-read support by variant class

ALT reads per variant: all records



	SNV	insertion	deletion
0	42,140	922	1,069
1	917	19	16
2-4	487	10	7
5-9	144	4	1
10-49	204	3	2
50-199	83	2	1
200+	30	1	2

ALT reads per variant: positive support only

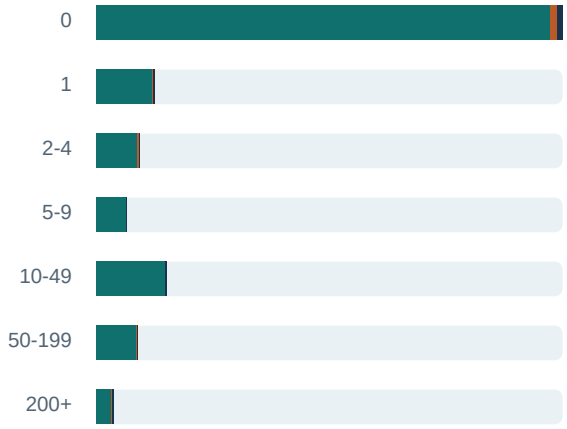


	SNV	insertion	deletion
1	917	19	16
2-4	487	10	7
5-9	144	4	1
10-49	204	3	2
50-199	83	2	1
200+	30	1	2

The positive-support panel excludes zero-ALT records. Bars stack SNV, insertion, deletion and other-replacement record counts. Matching colored columns print each class count directly. These are variant counts, not read counts.

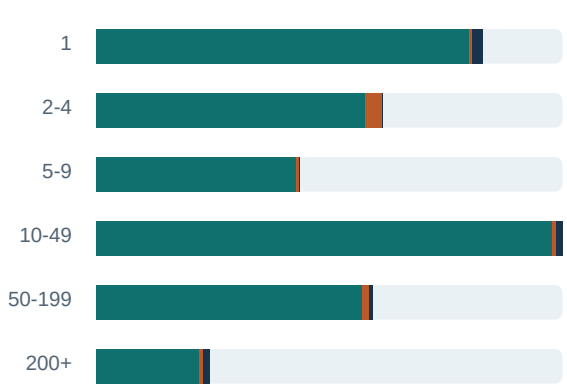
RNA ALT reads: exon-involved small variants

ALT reads per exon-involved variant: all records



	SNV	insertion	deletion
0	869	14	10
1	108	1	3
2-4	78	5	0
5-9	58	1	0
10-49	132	1	2
50-199	77	2	1
200+	30	1	2

ALT reads per exon-involved variant: positive support only



	SNV	insertion	deletion
1	108	1	3
2-4	78	5	0
5-9	58	1	0
10-49	132	1	2
50-199	77	2	1
200+	30	1	2

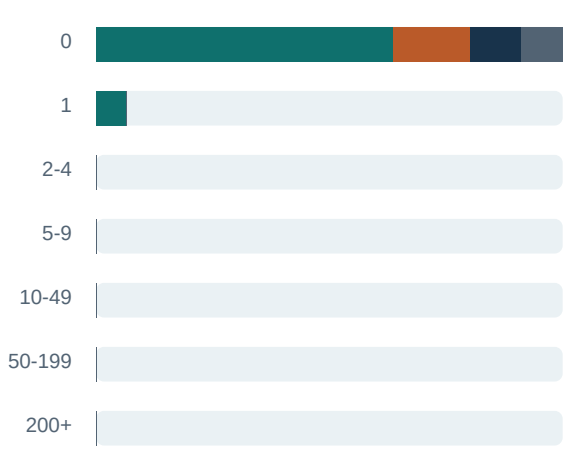
The input is the exon-involved subset of the small-variant source calls. The first panel retains zero-ALT calls; the second conditions on positive ALT reads. Insertion flanks are candidate exon involvement. SVs have a separate plot because a split alignment is not an ALT-spanning read.

Exon-involved nominated SVs: RNA breakpoint screen

97 of 188 nominated SV source records involve an Ensembl 87 exon. Intrachromosomal DEL/DUP/INV records query their full nominated span; INS queries insertion flanks; TRA/BND query nominated breakpoints. 4 records have insufficient breakpoint coordinates to rule out exon involvement and are excluded from this plot.

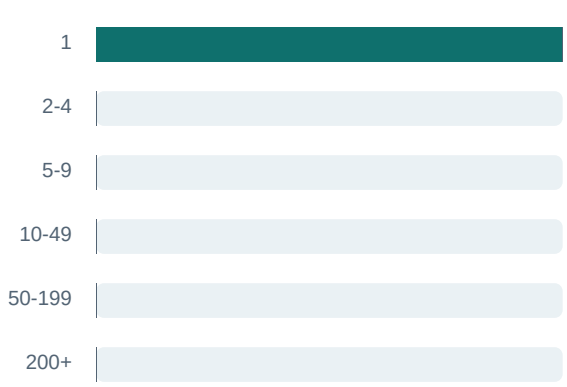
RNA screen	Exonic SV records	Candidate reads or pairs
Same-read split alignments	0	0
Cross-end primary mates	6	6

RNA link candidates per exon-involved SV: all records



	DEL	DUP	INV	BND
0	58	15	10	8
1	6	0	0	0
2-4	0	0	0	0
5-9	0	0	0	0
10-49	0	0	0	0
50-199	0	0	0	0
200+	0	0	0	0

RNA link candidates per exon-involved SV: positive support only



	DEL	DUP	INV	BND
1	6	0	0	0
2-4	0	0	0	0
5-9	0	0	0	0
10-49	0	0	0	0
50-199	0	0	0	0
200+	0	0	0	0

Bars count SV source records by reported class, binned on split-read plus cross-end mate candidate counts. Same-read supplementary alignments and opposite primary mates are different kinds of screening evidence; neither validates a junction sequence or translated product. The screen uses 500-base windows and mapping quality at least 20. The COLO829 benchmark call sets remain separate in the underlying table.

Original-read replay: junction versus RNA path

This selected-event replay uses oriented DNA adjacencies and actual RNA alignment records, including soft-clipped bases and low-MAPQ pieces. Direct junction fragments count reads crossing the same RNA join, even when their wider transcript sequences differ. MAPQ 20 is a sensitivity subset, not a gate on whether the junction sequence was observed. Isovar extends these joins through overlapping original read sequences; one read need not span the assembled context. Context length is a bounded assembly result, not an isoform-support count.

Nominated DNA record	Class	RNA join, all MAPQ	Join, MAPQ 20 subset	Longest context (bp)
SV3_DEL **CNIH3** chr1:224595102-224595102 chr1:224598323-224598323	DEL	3	0	1213
SV42_DEL **CNOT1** chr16:58590638-58590638 chr16:58629419-58629419	DEL	0	0	0
SV49_DEL **LARGE1** chr22:33363264-33363264 chr22:33442413-33442413	DEL	0	0	0

Gene names overlap the source breakends (Ensembl 87); they do not establish a coding fusion. IGV loci are one-based inclusive, GRCh38, with no digit separators.

The chr1 SV3 deletion has six opposite-mate candidate pairs and three original fragments with an identical split junction sequence containing 19 unplaced bases. Its linked alignment pieces span MAPQ 1-52, including low-MAPQ supplementary pieces; the MAPQ 20 subset loses all three joins. Assembly extends this sequence to 1,213 bases through overlapping reads and reaches an extension limit. SV42 and SV49 have one mate candidate each but no directly observed join. SV3 is outside the exon-involved funnel above.

The HCC1395 replay BAM is a source-derived 309-record regional fixture; COLO829 uses the 129,534-record selected-site BAM. Both were extracted after aligning the complete original RNA run. No transcript models were supplied in this geometry audit, so frame and SV-derived protein status remain unresolved. Other splicing and DNA-to-RNA causation have not been excluded.

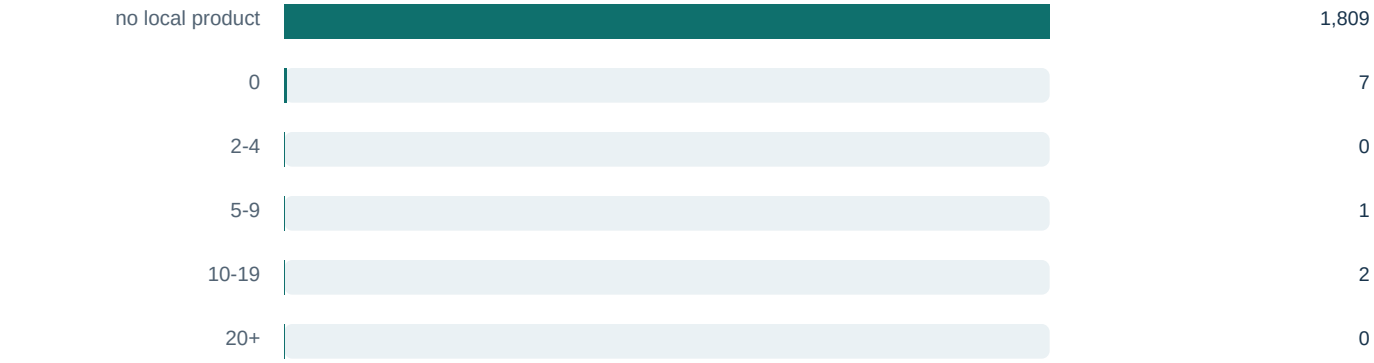
Isovar-marked amino-acid span

For each RNA ALT-observed call, count amino-acid positions in the mutation interval of its top-ranked local protein context. This is not an independently verified count of changed residues: synonymous DNA calls can produce a reference-identical amino acid window. Calls with no local context are shown explicitly. The second panel omits the one-residue category to expose the longer tail; its denominator is smaller. A zero-length interval can represent a deletion junction. These windows do not determine a complete frameshift tail or a full-length isoform.

Marked AA positions: all ALT-observed calls



Marked AA positions: excluding one-AA calls



Highest-support SNVs with local coding context

Top 3 of 120 ALT-supported calls in this category, ranked by source-library ALT fragments. Window support is reported separately. A reference-identical sequence is a coding context, not a neoORF. DNA effect labels and the displayed reference may use different transcript isoforms.

TUBB2A | chr6:3153911:G>A

IGV (GRCh38): chr6:3153911-3153911 | Silent p.A430= | 677 ALT fragments

564 reported window fragments; 26 aa. Transcript(s): ENST00000333628

REF	MEFTEAESNMNDLVSEYQQYQDATAD
RNA	MEFTEAESNMNDLVSEYQQYQDATAD

Reference-identical local context. Reference: ENST00000333628, residue 406. No unobserved sequence is filled from reference.

ETV5 | chr3:186048838:T>C

IGV (GRCh38): chr3:186048838-186048838 | Substitution p.Y445C | 584 ALT fragments

299 reported window fragments; 32 aa. Transcript(s): ENST00000306376;ENST00000434744

REF	VAGERYVYKFVCDPDALFSMAFPDNQRPFLKA
RNA	VAGERYVCKFVCDPDALFSMAFPDNQRPFLKA

Differing focal RNA residues highlighted. Reference: ENST00000306376, residue 438. No unobserved sequence is filled from reference.

PLD3 | chr19:40374523:C>T

IGV (GRCh38): chr19:40374523-40374523 | Silent p.L308= | 578 ALT fragments

211 reported window fragments; 28 aa. Transcript(s): ENST00000356508;ENST00000409281;ENST00000409419;ENST00000409587;ENST00000409735

REF	SGRTPDLKALLNVVDNARSFIYVAVMNY
RNA	SGRTPDLKALLNVVDNARSFIYVAVMNY

Reference-identical local context. Reference: ENST00000356508, residue 302. No unobserved sequence is filled from reference.

Top RNA-supported frameshift nominations

Top 3 of 4 ALT-supported calls in this category, ranked by source-library ALT fragments. Window support is reported separately. A reference-identical sequence is a coding context, not a neoORF. DNA effect labels and the displayed reference may use different transcript isoforms.

CDKN2A | chr9:21971154:CCG>C

IGV (GRCh38): chr9:21971154-21971156 | FrameShift p.A68fs | 6,967 ALT fragments

5,248 reported window fragments; 24 aa. Transcript(s): ENST00000304494;ENST00000479692;ENST00000494262;ENST00000497750;ENST00000498124;ENST00000498628;ENST00000578845;ENST00000579122

REF	**MMMGSARVAELLLLHGAEPNCADP**
RNA	**MMMGSARVAELLLLHG**GAQLRRPR

Differing focal RNA residues highlighted. Reference: ENST00000304494, residue 52. No unobserved sequence is filled from reference.

FHL2 | chr2:105361307:G>GCCCC

IGV (GRCh38): chr2:105361307-105361307 | FrameShift p.C388fs | 845 ALT fragments

654 reported window fragments; 35 aa. Transcript(s): ENST00000322142;ENST00000344213;ENST00000393352;ENST00000393353;ENST00000408995;ENST00000409177;ENST00000409807

REF	**FNCKKCSLSLVGRGFLTERDDILCPDCGKDI**
RNA	**FNCKKCSLSLVGRGFLTERDDIL**WGPRLWERHLNS

Differing focal RNA residues highlighted. Reference: ENST00000322142, residue 249. No unobserved sequence is filled from reference.

NBPF11 | chr1:148105470:C>CATCT

IGV (GRCh38): chr1:148105470-148105470 | FrameShift p.V788fs | 3 ALT fragments

2 reported window fragments; 21 aa. Transcript(s): ENST00000614015;ENST00000614785;ENST00000615281

`LQDSLDRCYSTPSSCLEQPDS`
No unambiguous pinned reference-window placement; comparison remains unresolved.

Top other insertions, deletions and replacements

Top 3 of 64 ALT-supported calls in this category, ranked by source-library ALT fragments. Window support is reported separately. A reference-identical sequence is a coding context, not a neoORF. DNA effect labels and the displayed reference may use different transcript isoforms.

DDX17 | chr22:38489856:GA>G

IGV (GRCh38): chr22:38489856-38489857 | Intronic intronic | 221 ALT fragments

No annotated coding reference context; the original run stopped before RNA assembly. The following DDX17 deep dive finds intronic coverage and alternative splicing, but no ALT-phased coding path. No protein sequence is inferred.

GOLGA5 | chr14:92811641:GAGA>G

IGV (GRCh38): chr14:92811641-92811644 | Deletion p.K404del | 109 ALT fragments

85 reported window fragments; 33 aa. Transcript(s): ENST00000163416

REF	LAERKYSDEKKRVDELQQQVKLYKLNLESSKQEL
RNA	LAERKYSDEK|RVDELQQQVKLYKLNLESSKQEL

Verified in-frame deletion; the vertical marker shows the missing reference residue(s). Reference: ENST00000163416, residue 395. No unobserved sequence is filled from reference.

Deleted reference interval: 405-405 (one-based). The original atlas effect label has a one-residue numbering error (Varcode #643); the aligned sequence interval is independently verified.

TMED8 | chr14:77337393:A>AT

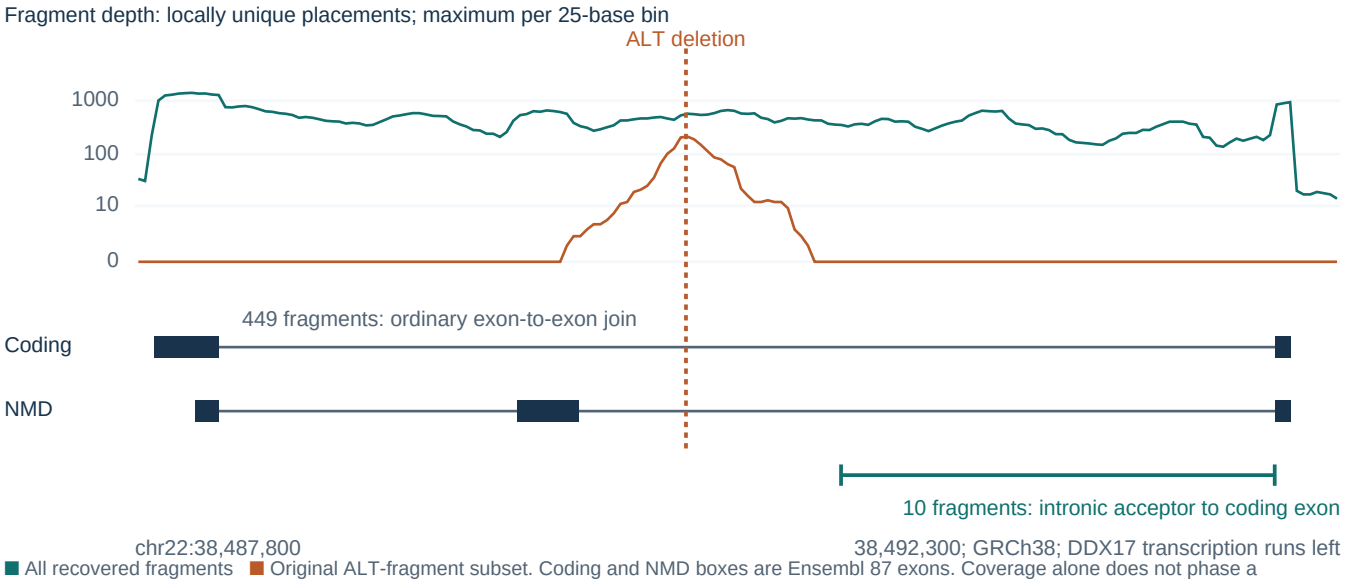
IGV (GRCh38): chr14:77337393-77337393 | ThreePrimeUTR 3' UTR | 61 ALT fragments

RNA ALT observed, but no local protein context was reconstructed. No protein sequence is inferred.

DDX17: intronic ALT RNA, alternative splicing and the missing protein

chr22:38489856:GA>G | IGV (GRCh38): chr22:38489856-38489857 | 221 original ALT fragments. The old protein run found zero annotated coding reference contexts and returned before RNA assembly. It did not test whether cryptic exons or retained introns existed.

The follow-up recovers both mates for 603 selected fragments and expands to 27,309 pairs from the same SRR8615617 library. Splice-aware STAR alignment to a 50-kb DDX17 reference finds continuous intronic coverage (minimum 115 fragments), ordinary exon joins, NMD-model joins and alternative intronic acceptors.



Evidence / GRCh38 interval	Result and interpretation
Ordinary join: chr22:38488116-38492055	449 locally unique fragments; this splice removes the mutation-containing intron.
Alternative join: chr22:38490437-38492055	10 locally unique fragments; canonical minus-strand splice motif; no direct ALT-fragment linkage.
ALT-linked alternative: chr22:38486940-38489622	One ambiguous fragment; 9-base anchor, two local placements, noncanonical motif, neither end an annotated exon.
All 221 original ALT fragments	214 locally unique, 5 multiply placed, 2 unaligned in the expanded local run; no locally unique ALT-fragment splice.

Why no defensible mutant protein yet? The deletion is intronic in both complete coding models; a retained-intron annotation covers it but lacks a complete CDS. Continuing either annotated coding frame into the retained intron predicts a stop at **chr22:38492042-38492044**. The 10-fragment alternative path predicts a stop at **chr22:38490360-38490362**. Both stops precede the deletion in the minus-strand translation direction. These are reference-path predictions, not ALT-phased proteins; cryptic initiation and other paths remain unresolved.

Scope: SRA Toolkit 3.4.1; STAR 2.7.11b without supplied splice annotations; GRCh38 local sequence verified against all four Ensembl 87 transcript sequences. The regional SRA index uses GRCh37; recovered reads are realigned to GRCh38. Local uniqueness is not genome-wide uniqueness. This bounded selection cannot establish whole-library absence or mutation causality. Original pairs, every alternative placement, source coordinates, hashes and commands: ddx17_audit/manifest.json and README.md. Reusable intronic assessment remains tracked in #583.

SV-derived RNA transcripts: junctions, branches and unresolved frames

These are sequence-supported RNA paths. A breakpoint-associated transcript is not automatically a coding fusion or a DNA-causal product. Split placement, soft clips, original query sequence and exon anchors are complementary evidence.

CNIH3 | COLO829_SOMATIC_SV3_DEL

IGV (GRCh38): chr1:224595102-224595102 chr1:224598323-224598323

3 original junction fragments; 1 pinned example RNA paths; representative local-window support 3 fragments. These measures are not pooled.

RNA junction context (nucleotides; no protein frame assigned):

AATTTGCCCTGTTGTCAGGCCTATAGTAGAACTTCTTTTGGAGTTGCTTCTTATGGATG
The newer shared SV3 audit assesses all retained paths and original witnesses: 3 placed junction fragments and 5 exact junction-window fragments in the expanded same-library source. Exon-linked sequence lies 280 nt downstream; no annotated frame is resolved. Split/mate-only screening missed this evidence.

Coding-fusion exemplar

No event-linked annotated-frame protein was resolved in the new COLO829 inventory replay. SV3 exploratory ORFs are not promoted to coding proteins.

ncORFs and other antigen-source categories

2 ALT-supported calls → 3 validated ncORF models → 1 amino-acid-changing models at default coverage. The separate single-read sensitivity pass adds 0 low-support model(s). All input models are assessed; ordinary CDS competitors remain in the exported results. Alternative ORFs on one call reuse its molecules and are not separate mutations.

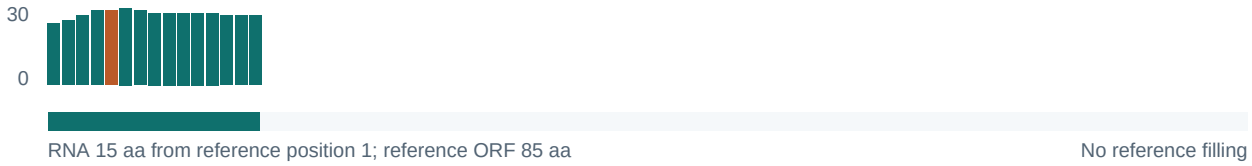
Gene / IGV (GRCh38)	Reference ORF	ALT	RNA result
DYNLL2 chr17:58089866-58089866	c17norep133	45	Changed local protein; 30 assembled fragments
DYNLL2 chr17:58089866-58089866	c17riboseqorf123	45	Focal allele is synonymous in this ORF
PITPNM1 chr11:67505274-67505274	c11riboseqorf96	11	Focal allele is synonymous in this ORF

DYNLL2 | c17norep133 | partial RNA coding candidate

30 assembled fragments; 21 fragments each span the displayed coding interval; minimum base depth 24. SRR8615617; default coverage.

REF	**MPSLGALRCVCLWCV**
RNA	**MPSLEALRCVCLWCV**

RNA fragment support per codon (minimum base depth; stop included when observed)



Complete reference ORF (RNA above is not extended from this sequence):

```
MPSLGALRCVCLWCVGVGSRFVLLSLSEFLSKGCVFVDYLLLFFLPEIMTGGCWGRISLNLFFLYIFLLS
APCLALSFLIPLEFS
```

The observed start-to-residue-15 window has G5E relative to this 85-aa downstream ORF. A longer 23-aa candidate retains 28 assembled fragments but only 11 single-read spans (minimum base depth 15). The overlapping c17riboseqorf123 frame and PITPNM1 uORF are synonymous for their focal alleles; they are not counted as mutant proteins.

■ Focal change ■ Unclassified nonfocal difference ■ RNA-covered interval. * Observed stop. Reference residues stay neutral. Coverage outside the displayed RNA is unassessed.

Sources: ORFdata 7,264-model catalogue 2026-10-11.1; complete ORF paths/proteins validated against Ensembl 87. COLO829 uses the whole-run selected-site BAM. HCC1395 uses all 36 original ALT witnesses for these loci, recovered from SRA and realigned genome-wide with minimap2 2.26; all five ALT counts reproduce. Default minimum read coverage is 2; the sensitivity pass uses 1 and reuses the same molecules. Full sequences, model alternatives, original witnesses, source hashes and budgets: ncORF audit manifest.

Assessment limits: RNA and reference Ribo-seq/MS annotations do not measure mutant translation or HLA presentation. General cryptic initiation and retained-intron ORFs remain unresolved beyond the focused DDX17 audit. Viral RNA/ORFs and cancer/testis isoform surveys are not assessed. Germline/co-somatic colors require source-qualified genotype and explicit phasing.

Sources, processing and limits

All RNA libraries, DNA nominations, polished inputs and references across the report set are listed in SEQUENCING_SOURCES.md. Focused CDKN2A and EIF3K-CYP39A1 reports show highlighted reference alignments and original-RNA support separately from these atlas summaries.

Input	Pinned scope
Matched-normal small variants	SMaHT COLO829/COLO829BL, GRCh38 no-alt
SV benchmarks	Paulin Paz 2024 v2 (54 records) and Valle-Inclan 2022 lifted (134 records)
Original RNA	SRR8615617; 115,269,780 paired Illumina spots / 230,539,560 reads
Alignment	minimap2 2.26 -ax sr --frag=yes; NCBI GRCh38 no-alt

Reference model and interpretation

Small-variant coding effects use Varcode with Ensembl release 87 on GRCh38. The human orfdata 0.34.1 catalog uses Ensembl 101 GRCh38 reference ORFs (7,264 models, dataset 2026-10-11.1). The new ORFdata 0.39.1 replay validates nine complete catalogue ORF paths/proteins against Ensembl 87, then reconstructs their frames on mutation-supporting RNA through the shared library API. The ncORF chapter separates default and single-read results. Genomic transcript overlap is not RNA-compatible isoform support. Original-read allele counts and local protein contexts are separate observations.

SHA-256 input pins

SMaHT SNV VCF gzip

`e57f77c8bc280bbad2255856c4062348f92f25da38726f367322c52736578f31`
SMaHT indel VCF gzip

`e8d0fb67925a2365b485cd65feab8dfdc12bbe54c0bc0bdec44c1846b518b685`
SV v2 GRCh38 VCF

`8ad075bf43aa71d89c591396569e813959ada44a30f171982c2d299411290bac`
Earlier SV lifted VCF

`a8f38454367cf8837a7dba1d29c9bc338a8e62cadfee6d2489931f8105f0dcdc`
SRR8615617 SRA archive

`9d89bb4f5a74fba7a8ba9495fd44046c35f4abb60d10c7bd9261f07569520b4f`
GRCh38 no-alt FASTA (uncompressed)

`9cce8b926416dd96b152deea85188495b75f7ac8d634cc723a017067be8702b7`
orfdata bundled GRCh38 catalog

`06bc0bf3f81c25a7356f71a38581ff5704c05dcb75752454bc1ca75075ab8c09`
Ensembl 87 GRCh38 protein FASTA gzip (self screen)

`d873c199d5dbf20869db3f2cbd6ac726259215d91c2782f9f1ddcad4bb117cf9`

What remains unresolved

The SMaHT DNA, CCLE RNA and TAGET polished isoforms are separate cultures. The short-read genomic alignment is not splice-aware; junction-crossing reads may be lost. Three sequence-supported SV3 junction fragments have low-MAPQ linking pieces; placement, splice alternatives, full-length isoform linkage and SV-derived proteins remain unresolved.

Machine-readable per-call annotation, allele, protein, and SV tables and complete reconstruction commands are published with the Isovar HCC1395/COLO829 reports. These PDFs summarize selected public source call sets, not a complete catalogue of every cell-line mutation.

COLO829 | CDKN2A frameshift

High-support local mutant protein window - not a complete mutant protein

GRCh38 chr9:21971154 CCG>C. A two-base deletion changes the coding frame in the Ensembl 87 CDKN2A reference shown below (ENST00000304494 / ENSP00000307101).

IGV (GRCh38): chr9:21971154-21971156

RNA support: allele and complete window are different measurements

6,967 ALT fragments / 6,984 ALT allele-read observations

5,248 fragments support this entire 24-aa window

The window has 5,338 supporting allele-read observations. The locus also has 551 REF fragments and 252 other-allele fragments. These are template counts in one Illumina library, not UMI-confirmed molecules.

Reference versus observed mutant window

Reference residues 52-75; the altered tail starts at reference residue 68.

Reference 52-75	MMMGSARVAELLLLHGAEPNCADP
RNA mutant	MMMGSARVAELLLLHGGAQLRRPR
Change	^^^^^^^^

The highlighted mutant tail begins after a 16-aa shared prefix; altered tail: 8 aa (GAQLRRPR). Columns after the change are boundary-aligned, not a homology alignment. No stop or full-length mutant protein is resolved by this local window.

Retained alternatives - never add their support

Rank 1: 24 aa, 5,248 complete-window fragments

Rank 2: 23 aa, 5,333 complete-window fragments

Rank 3: 22 aa, 5,354 complete-window fragments

1,416 overlapping local contexts remain in the atlas. Eight transcript IDs are compatible with the displayed window; this does not select a unique isoform. All ALT reads were evaluated by the recorded high-depth read-spanning method.

Interpretation

This is a strongly RNA-supported frameshift window. RNA support does not establish translated protein abundance, tumor specificity in another culture, or antigen presentation.

COLO829 | CDKN2A frameshift

Reference protein, observed mutant window, and assembly extent

The full reference protein is shown for orientation. Only the 24-aa RNA window is resolved in this report; dots in the mutant row mean unobserved sequence, not gaps filled from reference. Coordinates are anchored to the reference before the frameshift.

Reference coordinates 1-60

Reference	MEPAAGSSMEPSADWLATAAARGRVEEVRALLEAGALPNAPNSYGRRPIQVMMMGSARVA
RNA mutant	.....MMMGSARVA

Reference coordinates 61-120

Reference	ELLLLHGAEPNCADPATLTRPVHDAAREGFLDTLVVLHRAGARLDVRDAWGRLPVDLAEE
RNA mutant	ELLLLHGGAQLRRPR.....

Reference coordinates 121-156

Reference	LGHRDVARYLRAAAGGTRGSNHARIDAAEGPSDIPD
RNA mutant	.....

Assembly / reference annotation track



Reference only RNA observed Altered tail

RNA coverage is restricted to reference-anchored positions 52-75 in this selected window, supported by 5,248 reported fragments. The boundaries at 52 and 76 mark the observed window, not validated full-ORF ends. No reference sequence is stitched into the mutant sequence. Upstream initiation and downstream stop remain unresolved.

Variant provenance

Focal change Germline Phased somatic

No additional non-reference residues occur in the displayed RNA sequence relative to its indicated reference contributor(s). No germline or other somatic allele is assigned without genotype and phasing evidence. Missing sequence stays unobserved.

COLO829 | CDKN2A frameshift

All inputs used in this focused report

Original RNA: SRR8615617 / PRJNA523380 / SAMN10988208

COLO829 CCLE Illumina paired cDNA RNA-seq: 115,269,780 pairs, 101 bases per end. The whole run was aligned to GRCh38 before selected-site extraction. The atlas BAM is a derivative of this same library, not independent evidence.

DNA nomination: SMaHT COLO829BLT50 reference set

Truthset_Indel_COLO829.vcf.gz; PASS, COLO829T-origin call relative to COLO829BL. DNA and RNA cultures come from different studies.

Genome and protein reference

NCBI GRCh38 no-alt analysis FASTA; Ensembl 87 transcript ENST00000304494. Pinned reference window starts at residue 52. Protein SHA-256: 7a2d16efe0bed2c12560ef3a9b1678b1f7aff329bbb1cf55ad9f8d826f5ed98a

Processing and exact original witnesses

minimap2 2.26 -ax sr --frag=yes (genomic, not splice-aware); Isovar 1.59.6 atlas, default read handling and high-depth direct contexts. The call, protein and ALT-witness tables retain the source-run IDs. Re-rendering does not rerun the atlas.

Complete study/source index and pinned inputs

SEQUENCING_SOURCES.md lists all sources across the three-sample reports. The adjacent COLO829_CDKN2A_frameshift.json pins the call, reference, sequence and table SHA-256 digests.

Compatible transcript IDs

ENST00000304494, ENST00000479692, ENST00000494262, ENST00000497750, ENST00000498124, ENST00000498628, ENST00000578845, ENST00000579122